Research Statement

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How can imperfect quantum resources be transformed into reliable computational advantages?

Research Identity: From Software Systems to Quantum Information

This question has guided the development of my undergraduate research. I entered quantum information from Software Engineering rather than through a conventional physics curriculum. My initial training taught me to decompose complex systems into algorithms, interfaces, data structures, and verifiable computational components. It also made me sensitive to efficiency, robustness, reproducibility, and implementation cost. Through a supervisor-led reading group, I began studying quantum states, circuits, algorithms, cryptographic protocols, quantum key distribution, and quantum communication. I reinforced this theoretical preparation through Python, Qiskit, and PennyLane simulations. Over time, I moved from learning established models to identifying structural weaknesses, proposing mathematical abstractions, and building computational workflows to test them. This progression shaped my research identity: I am interested not simply in ideal quantum advantage, but in the mechanisms by which limited, noisy, heterogeneous, or imperfect quantum resources can be organized into secure and useful systems.

Protocol Structure: When Does a Quantum Resource Actually Provide Security?

My first sustained attempt to answer the central question began with quantum signatures. The motivating problem was structural: a protocol may be built from legitimate quantum primitives and still be insecure because those primitives interact incorrectly at the system level. In our work on Grover-based quantum signature schemes, we analyzed how publicly derivable verification operations and exposed communication structure could permit key recovery, forgery, and repudiation. The lesson was that quantum key distribution, teleportation, or Grover operations do not automatically transfer their security guarantees to a larger protocol. Security depends on composition: authentication structure, information flow, verification authority, and the separation of public and private resources.

My contribution was not limited to checking individual attacks. I helped organize the mathematical framework and notation, supported the abstraction of four reusable inspection points for protocol analysis, implemented Qiskit-based verification, and contributed to the strengthened arbitrated signature construction and manuscript preparation. This work gave me a first answer to the central question: imperfect or incomplete protocol structure cannot be compensated for by invoking powerful primitives. Reliable advantage requires a system-level design in which every resource has a clearly defined security role. That conclusion led to the next question: if quantum resources are limited, how should they be allocated so that an improvement remains meaningful after cost is considered?

Finite Resources: How Should Quantum Information Be Allocated and Evaluated?

I explored this question through an error-corrected, quantum-Huffman-inspired probabilistic source-decoding framework. The aim was deliberately narrower than constructing a strictly lossless quantum Huffman code. Instead, I asked whether classical source statistics could guide finite-dimensional quantum state design without requiring coherent variable-length transmission. I reinterpreted Huffman lengths as support budgets in a common Hilbert space, then introduced group ancilla to transfer inter-group ambiguity into orthogonal label information while leaving smaller conditional state-discrimination problems.

The originality of this work lies in connecting source structure, quantum state geometry, probabilistic decoding, and explicit resource accounting. The framework combines support allocation, group-conditioned POVMs, effective error correction, MAP posterior decisions, and resource-normalized margins. I developed analytical sufficient conditions and SDP-based certificate logic for determining when a grouped decoder can outperform a fixed-length baseline under a stated cost convention. Equally important, I treated negative results as informative. Representative benchmarks showed that advantage is not generic when ancillary and error-correction overhead dominate, whereas favorable-regime sweeps and exact small-instance certificates identified structured regions in which support savings, ambiguity transfer, and low overhead align.

This project gave me a second answer to the central question: a computational advantage is reliable only when the gain survives explicit resource normalization. Additional ancilla, redundancy, or algorithmic sophistication cannot be treated as free. Yet the work also suggested a broader possibility. If limited resources can be reshaped through support geometry and conditional discrimination, perhaps imperfect entanglement links in a network need not be treated only as bottlenecks. This raised the next question: can weak entanglement itself become a functional resource?

Weak-Link Communication: Can Imperfection Be Designed Rather Than Eliminated?

Realistic quantum networks contain heterogeneous and noisy links. Long-distance channels, imperfect couplings, and mixed hardware platforms make uniformly strong entanglement unrealistic. Conventional approaches often treat weak links as defects to be distilled away, bypassed, or ignored. I instead investigated whether weak entanglement could be used as a controllable task-level resource.

In my weak-link entanglement-assisted no-signaling communication framework, cooperating groups are connected by sparse inter-group links. The leading network contribution is modeled through link visibility, local group bias, measurement-basis alignment, and a sign strategy. This produces complementary first-order corrections for AND-type coordination tasks and OR-type fault-tolerant tasks. I further developed a unified mapping from Werner, depolarizing, phase-damping, and amplitude-damping noise to effective correlation parameters, together with Bayesian reliability weighting and experimentally scannable performance regions. I also examined how the same analytical variables could be calibrated across photonic, trapped-ion, and superconducting platforms.

The key conceptual contribution was to change the role assigned to weak links. Instead of asking only how much performance they destroy, I asked how their strength, orientation, and topology can be optimized for different objectives. Weak entanglement becomes useful when it is embedded in a task-aware architecture. This provided a third answer to the central question: imperfection can become a computational resource when its effect is parameterized, calibrated, and aligned with system goals. Once weak links were represented as network edges with task-dependent roles, however, a new problem emerged: how should an entire network coordinate its behavior when cooperation, competition, topology, and noise all change together?

Quantum Networks and Games: How Can Distributed Systems Make Adaptive Decisions?

This question led to my first-author work on weak-link multi-group quantum games. I proposed a graph-based framework in which cooperative groups form the vertices of a weak-link interaction network, while heterogeneous inter-group resources are encoded by weighted signed edges. The leading effect is summarized through a game tensor that links topology, group bias, task orientation, and weak-link strength. The analytical layer derives task-dependent sign rules, first-order advantage conditions, spectral bounds, and bounded-degree scalability estimates.

The project then extends beyond a static game model. I relaxed discrete signs and group biases into differentiable variables and introduced a learnable quantum-game layer optimized through a hybrid quantum-classical workflow. I implemented automated search, batch evaluation, repeated-trial analysis, topology and noise comparisons, visualization pipelines, and backend-side circuit checks. In selected surrogate experiments, learned structures remained aligned with analytical sign rules while adapting to topology and noise. I also made a strict distinction between theorem-level quantities, surrogate objectives, and backend proxy measurements, because optimization should support analytical understanding rather than replace it.

This work connected my earlier interests in weak-link communication, graph structure, resource accounting, and software implementation. It also changed my view of a quantum network. A network should not be treated merely as a passive channel that transfers states or keys; it can be modeled as a strategic system that allocates resources, coordinates agents, and adapts its decision policy. This provided a fourth answer to the central question: reliable computational advantage can emerge from imperfect network resources when analytical structure and adaptive optimization are combined.

Toward Quantum Machine Intelligence

The natural continuation of this research is quantum machine intelligence: systems in which quantum-network structure, measurement choices, routing policies, and task objectives can be learned while remaining interpretable. My previous work suggests that learning is most useful when it is constrained by analytical models. Purely black-box optimization may produce high surrogate scores without clarifying whether the result reflects a physical task value, a numerical artifact, or an implementation-specific proxy. I therefore want to study hybrid methods in which theoretical guarantees, graph structure, convex certificates, and calibrated noise models define the space within which learning operates.

From this perspective, machine intelligence is not an unrelated addition to quantum communication. It is the mechanism that may allow imperfect quantum systems to adapt. A future network will need to decide which links to activate, how to distribute entanglement, which basis or protocol to use, when redundancy is worthwhile, and how to balance security, throughput, and computational cost. These decisions are naturally sequential, graph-structured, and resource constrained. My goal is to develop models in which such decisions can be learned without sacrificing physical interpretability or rigorous evaluation.

Future Research: A Resource-Aware Quantum Internet

My future research plan is to extend the central question toward the architecture of a resource-aware Quantum Internet. I am particularly interested in three connected directions. First, I want to develop quantum-network protocols that jointly optimize topology, routing, entanglement quality, security, and task performance under finite-resource constraints. This includes models for heterogeneous links, distributed entanglement allocation, adaptive routing, and end-to-end task-aware performance rather than isolated channel metrics.

Second, I hope to study learnable quantum-network models in which graph structure, measurement choices, and communication policies adapt under calibrated hardware noise. My objective is not only to optimize a surrogate score, but also to preserve analytical guarantees, identify the origin of any apparent advantage, and distinguish physical performance from proxy quantities. Techniques from graph learning, variational optimization, Bayesian decision theory, and convex certification may provide a unified methodology.

Third, I want to investigate distributed quantum computing and quantum cryptography as components of one architecture rather than separate layers. Security, communication, computation, and error correction all consume the same limited resources. A protocol that is secure but too expensive, or efficient but structurally vulnerable, cannot support a practical network. I therefore hope to develop frameworks in which these objectives are evaluated together.

These directions converge on my central research vision: the future Quantum Internet should not depend on perfect resources. It should be secure, adaptive, interpretable, and capable of extracting useful computation from imperfect devices and heterogeneous links.

Preparation and Long-Term Objective

My background in Software Engineering provides practical tools for this agenda. I am comfortable with Python-based scientific computing, numerical experiments, data analysis, visualization, Qiskit and PennyLane workflows, LaTeX manuscript preparation, and reproducible research organization. Mathematical modeling competitions and non-quantum software projects have trained me to translate real problems into computational models and to test assumptions through simulation. At the same time, I recognize the need to deepen my foundations in quantum information theory, optimization, cryptography, and distributed quantum systems through graduate training.

I am seeking a research-oriented MPhil or PhD environment in which rigorous theory, computational verification, and collaborative implementation are treated as complementary. I hope to contribute my experience in protocol modeling, graph-based analysis, scientific software, and manuscript development while learning from researchers working on quantum communication, quantum networks, quantum algorithms, quantum cryptography, or quantum machine learning.

My long-term objective is to become an independent researcher developing intelligent and resource-efficient quantum information systems. I do not regard resource limitations as obstacles to be removed before meaningful computation can begin. I regard them as structures that can be modeled, allocated, and exploited. The question that began my undergraduate research will therefore continue to guide my graduate work: how can imperfect quantum resources be transformed into reliable computational advantages?